

Module 2 – Git Workflows for CI/CD

Why Git Workflows Matter in CI/CD

CI/CD pipelines are triggered by **Git events**: - push - pull request - merge

If your Git workflow is weak: - Pipelines run on wrong code - Bugs reach main branch - Releases become risky

A good Git workflow = stable CI/CD.

1. Main Concepts You Must Know

Branches

- `main` / `master` → production-ready code
- `feature/*` → new work
- `bugfix/*` → fixes

Never work directly on `main`.

2. GitHub Flow (Industry Standard)

This is the most common workflow.

Steps: 1. Create feature branch from `main` 2. Make changes 3. Push branch 4. Create Pull Request (PR) 5. CI pipeline runs 6. Review + merge

feature/login → PR → CI passes → merge to main

3. Pull Requests & CI

A Pull Request: - Triggers CI automatically - Blocks merge if pipeline fails

Why this is powerful

- No broken code in `main`
- QA gets early feedback
- Teams trust releases

4. Protected Branches

Protected `main` branch usually requires: - Passing CI checks - At least 1 approval

This enforces discipline.

5. QA Perspective (Very Important)

As QA / Manager QA: - CI must run on PRs - Failed tests = blocked merge - QA helps define quality gates

CI is not optional — it is a gatekeeper.

Key Takeaways

- Git workflow controls CI/CD behavior
 - PRs are the heart of CI
 - QA benefits directly from strong Git discipline
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👉 Next: Complete `tasks.md` to practice Git workflows.